



# Ritul Jangir

M24CSA028

M.Tech Artificial Intelligence

Indian Institute Of Technology (IIT), Jodhpur

+91-7014571928

rituljangir7014@gmail.com

m24csa028@iitj.ac.in

Github | LinkedIn

## EDUCATION

| Degree/Certificate | Institute/Board                         | CGPA/Percentage | Year      |
|--------------------|-----------------------------------------|-----------------|-----------|
| M.Tech. (AI)       | Indian Institute of Technology, Jodhpur | 8.43            | 2024-2026 |
| B.Tech. (I.T)      | M.B.M University, Jodhpur               | 8.32            | 2019-2023 |
| Senior Secondary   | CBSE Board                              | 87.4%           | 2018      |
| Secondary          | CBSE Board                              | 91.2%           | 2016      |

## WORK EXPERIENCE

- Musikaar** Jun 2023 - July 2024  
*Software Engineer* Ahmedabad
  - Showcased proficiency in leveraging Python, Selenium, and Pytest to seamlessly automate office-based testing workflows, leading to a 60% increase in testing efficiency.

## INTERNSHIPS

- Siemens EDA (India) Pvt. Ltd** June 2022 - Aug 2022  
*Software Engineer Intern* Noida
  - The project involved the conversion of all functionalities provided by the GUI to be accessible and operable via the CLI.
  - Collaborated with senior SDE's with a team of 6 people on project.

## RESEARCH PROJECTS & PUBLICATIONS

### SynCrash: Zero-Shot Accident Detection and Localization in Traffic Surveillance Videos Accepted @ CVPR 2026 Workshop

- Developed a multi-stage accident understanding pipeline for CCTV traffic videos consisting of temporal localization, spatial impact prediction, and collision-type classification.
- Fine-tuned VideoMAEv2-Giant on CARLA synthetic accident data with metadata-aware embeddings and CCTV-specific augmentation strategies for robust sim-to-real transfer.
- Designed a physics-informed localization module combining YOLO object detection, trajectory estimation, motion reasoning, and collision heuristics for impact point prediction.
- Achieved a private leaderboard score of 0.40 and ranked among the top teams in the ACCIDENT @ CVPR 2026 Challenge under a zero-shot real-world evaluation setting.
- Technologies:** VideoMAEv2, YOLO, CARLA Simulator, PyTorch, Computer Vision, Multi-Object Tracking, Video Understanding

### CLARIF: Anatomy-Aware CT Rib Fracture Report Generation Under Review

- Proposed CLARIF, a clinically-aware framework for automated radiology report generation from 3D CT rib fracture volumes.
- Introduced centroid-based relative positional encoding and hierarchical anatomy-aware feature extraction to improve spatial reasoning across rib regions.
- Designed a Clinical Context-Aware Decoder (CCAD) integrating learnable pattern dictionaries, anatomical location encoders, severity-aware gating, and auxiliary clinical prediction heads.
- Implemented hard-negative contrastive learning to distinguish clinically plausible but incorrect reports, improving diagnostic consistency.
- Achieved state-of-the-art performance with ROUGE-L of 0.728, METEOR of 0.395, and superior clinical precision/recall on RibDataset and RibFrac benchmarks.
- Technologies:** PyTorch, Vision Transformers, CTViT, Medical Imaging, Contrastive Learning, NLP, Radiology AI

### OphEdit: Training-Free Text-Guided Editing of Ophthalmic Surgical Videos Oral Paper @ MedViD 2026, ECCV Workshop

- Developed the first training-free framework for text-guided editing of ophthalmic surgical videos while preserving anatomical consistency and temporal coherence.
- Designed a deterministic DDIM inversion pipeline with selective Attention Value (V) tensor extraction and conditional CFG injection for structure-preserving video editing.
- Introduced a second-order ODE integration strategy to reduce temporal artifacts and maintain realistic surgical instrument-tissue interactions.
- Validated the framework through expert ophthalmologist evaluation across multiple surgical phase transformations, outperforming adapted open-domain video editors.
- Technologies:** Diffusion Models, DDIM, CogVideoX, Video Editing, Generative AI, Medical Video Analysis, PyTorch

### Sketch-to-Color Image Generation

GitHub

- Built a diffusion-based image generation pipeline for translating sketch inputs into realistic color images.
- Integrated sinusoidal timestep embeddings and a UNet-style denoising architecture for iterative image refinement.
- Implemented an end-to-end encoder–diffusion–decoder framework supporting conditional image synthesis from sparse visual inputs.
- **Technologies:** Diffusion Models, UNet, PyTorch, Computer Vision, Generative AI

## TECHNICAL SKILLS

---

- **Programming:** C/C++, Python, SQL
- **AI/ML:** GenAI, Deep Learning, Machine Learning, NLP, RAG, Prompt Engineering, LLM-Finetuning
- **Frameworks & Tools:** PyTorch, LangChain, LangGraph, scikit-learn, Docker, Git, Django & REST APIs

## ACHIEVEMENTS

---

- 2025 GATE DA AIR 210
  - Secured an **R<sup>2</sup> score of 91.93** in the **Flipkart GRiDLock 2.0 Traffic Challenge** by designing a machine learning forecasting pipeline that accurately predicted future demand using only one day of historical demand data.
  - Secured a rank under 100 among over 10,000+ participants in the Amazon ML Challenge, a competitive data science competition hosted by Amazon.
-